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Sharad Institute of Technology
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An Autonomous Institute
Department of Computer Science and Engineering

A
Mini Project Synopsis
on

TruthLens: AI-Powered News & Claim Verification System

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that,

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Studying in SY Computer Science and Engineering have successfully completed the Mini Project Synopsis "TruthLens: AI-Powered News & Claim Verification System" under the guidance and supervision of [Guide Name] during the academic year 2025-2026 Semester IV. This is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

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ABSTRACT

TruthLens is an AI-powered news and claim verification platform designed to combat misinformation through automated, evidence-grounded investigations. The user submits either an article URL or raw claim text. The system validates the input, extracts and sanitizes the content via multi-stage anti-SSRF protections, and uses Google Gemini JSON-mode prompting paired with backend Pydantic validation to break the content into atomic, falsifiable claims. Evidence for each claim is retrieved concurrently from the Google Fact Check Tools API and the GNews API using asynchronous HTTP requests. A rule-based verification engine cross-references the claims against retrieved evidence, assigning one of four standardized verdicts: Supported, Contradicted, Mixed, or Insufficient Evidence. A deterministic mathematical scoring formula computes an explainable credibility index from 0 to 100 from a neutral baseline of 50, scaled by normalized verdict balance, claim coverage, evidence source quality, and publisher diversity. A neutral editorial summary strictly grounded in the verified evidence is then synthesized by Gemini, stating investigative limitations and confidence tiers. The system is built with an asynchronous FastAPI backend that mounts the HTML, CSS, and JavaScript frontend on a single address, reports "Insufficient Evidence" instead of speculating, and maintains a browser-based session ledger in local storage for auditability.

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INTRODUCTION

Misinformation spreads quickly online, and readers rarely have the time or tools to check each claim in a news article against reliable sources. Manual fact-checking is accurate but slow, while purely automated opinions from language models can be unreliable when they are not tied to evidence.

TruthLens is an AI-powered news and claim verification platform that addresses this gap. It pairs large language model reasoning with real-time news retrieval, registered fact-checks, and a deterministic mathematical scoring formula, producing transparent, audit-ready verification dossiers.

The user can submit either an article URL (protected by built-in anti-SSRF checks and HTML sanitization) or raw pasted claim text. The system breaks the content into discrete claims, gathers supporting and contradicting evidence from independent sources, computes a credibility score between 0 and 100, and presents a neutral summary together with the evidence behind it.

The project is implemented as an asynchronous FastAPI backend written in Python, mounting a responsive browser-based frontend, and is covered by 34 automated unit and integration tests. A mock fallback mode allows the whole system to be run and evaluated without third-party API keys.

PROBLEM STATEMENT

Fake and misleading news spreads faster than corrections, and ordinary readers cannot manually trace every claim in an article back to credible sources. Existing manual fact-checking is slow and cannot scale to the volume of modern digital media, whereas AI-only judgments can be unexplained, prone to hallucinations, and may speculate when no corroborating evidence is available.

There is a need for a system that can automatically analyze a news article or claim, retrieve independent evidence, and give a clear, explainable credibility result, while admitting uncertainty when evidence is missing and remaining safe against misuse such as server-side request forgery when URLs are submitted.

TruthLens directly targets this challenge by implementing an automated, evidence-grounded verification pipeline that cross-references claims against registered fact-checking databases and open-source wire archives before computing an auditable credibility score.

OBJECTIVES

1. To accept news input as an article URL or raw text, extracting clean content while enforcing anti-SSRF defenses and HTML tag sanitization.

2. To break complex news reports into atomic, falsifiable claims using Google Gemini JSON mode validated against strict Pydantic schemas.

3. To retrieve evidence concurrently from registered fact-checking desks (Google Fact Check Tools API) and news wire archives (GNews API) using asynchronous I/O.

4. To evaluate claim stances and assign standardized verdicts: Supported, Contradicted, Mixed, or Insufficient Evidence.

5. To calculate an explainable credibility score from 0 to 100 using a deterministic mathematical formula based on normalized verdict balance from a baseline of 50.

6. To generate a neutral editorial explanation grounded strictly in verified evidence, stating qualitative limitations alongside formula-derived confidence tiers (High, Moderate, Low).

7. To maintain an auditable, searchable, and exportable session ledger using browser local storage.

LITERATURE REVIEW

K. Shu, A. Sliva, S. Wang, J. Tang, & H. Liu. (2017). "Fake News Detection on Social Media: A Data Mining Perspective." ACM SIGKDD Explorations Newsletter, Vol. 19, No. 1, pp. 22-36 [1].

This survey explains the characteristics of fake news from psychological and social viewpoints and reviews detection methods based on news content and social context. The authors point out that fake news is intentionally written to mislead, which makes detection from content alone difficult and shows the need for external evidence.

S. Vosoughi, D. Roy, & S. Aral. (2018). "The Spread of True and False News Online." Science, Vol. 359, No. 6380, pp. 1146-1151 [2].

The study analyzed the spread of verified true and false news stories on Twitter and found that false news reached more people and spread faster than true news. This confirms that corrections arrive too slowly when verification is done only by hand, and motivates automated, rapid verification.

J. Thorne, A. Vlachos, C. Christodoulopoulos, & A. Mittal. (2018). "FEVER: a Large-scale Dataset for Fact Extraction and VERification." Proceedings of NAACL-HLT 2018 [3].

The FEVER dataset contains over 185,000 claims that are verified against evidence and labelled as supported, refuted, or not enough information. It established the standard approach of retrieving evidence for a claim first and then deciding a verdict, including an explicit "not enough information" outcome. TruthLens follows the same principle through its standardized "Insufficient Evidence" result.

Z. Guo, M. Schlichtkrull, & A. Vlachos. (2022). "A Survey on Automated Fact-Checking." Transactions of the Association for Computational Linguistics, Vol. 10, pp. 178-206 [4].

This survey describes automated fact-checking as a pipeline of claim detection, evidence retrieval, verdict prediction, and justification production. It stresses the importance of explainability. The TruthLens pipeline follows these stages, using claim extraction, parallel evidence retrieval, a verification engine, and a transparent score with editorial justification.

Research Gap

Existing work mostly focuses on isolated pipeline components or on opaque, black-box classification models. TruthLens bridges this gap by combining LLM-based claim extraction, asynchronous multi-source retrieval across registered fact-checks and live news wires, a rule-based verification engine, and an explainable, deterministic scoring formula within a unified, interactive web application.

METHODOLOGY

• **Requirement Gathering:** The verification goals, input modes (URL and pasted text), evidence sources, and expected outputs (a credibility score with explanation) were defined, along with security and testability requirements.

• **System Design:** The system was divided into a frontend and an asynchronous FastAPI backend. The backend is organized into API routes, core configuration and SSRF defenses, Pydantic v2 schemas, services (extraction, retrieval, verification, scoring, synthesis), and integration clients for Gemini, GNews, and the Fact Check API.

• **Development:** The backend pipeline was implemented in Python. The frontend pages (landing page, verification workspace, samples, ledger, how-it-works, and methodology) were built with HTML, CSS, and JavaScript, communicating with the backend through REST endpoints.

• **Testing:** A suite of 34 asynchronous unit and integration tests, run with pytest, covers SSRF defense, article scraping, claim extraction, evidence matching, scoring algorithms, and endpoint responses.

• **Deployment:** The application runs on a Uvicorn server. The FastAPI app also mounts the frontend, so the complete web application and interactive OpenAPI Swagger documentation are available from a single address (http://127.0.0.1:8000). API keys are supplied through environment variables.

• **Investigation Execution:** The user submits a URL or text. The backend validates the input, extracts atomic claims, retrieves evidence in parallel, and verifies each claim against the evidence found.

• **Scoring & Reporting:** A deterministic formula calculates the credibility score from 0 to 100 starting from a neutral baseline of 50.0. Each atomic claim is assigned an evidentiary weight and direction based on its verdict (Supported: +1.0 direction, weight 0.9–1.2; Contradicted: -1.0 direction, weight 1.0–1.3; Mixed: -0.25 direction, weight 0.8; Insufficient Evidence: 0.0 direction, weight 0.5). A normalized verdict balance is computed and modulated by claim coverage, an evidence quality factor, and a publisher diversity bonus. The score is clamped between 0 and 100 and mapped to qualitative brackets alongside a formula-derived confidence tier (High, Moderate, Low). Gemini synthesizes a neutral editorial explanation under strict grounding constraints, accepting the pre-computed score.

• **Audit & Improvement:** Each investigation is saved in the browser-based session ledger in localStorage with timestamp, claim breakdown, and sources, which supports review, search, and Markdown brief export.

ALGORITHM

Start

1. Receive input: article URL or pasted claim text.

2. If input is a URL, validate scheme (http/https only), resolve hostnames via DNS, and reject loopback (127.0.0.1, ::1), private RFC 1918, link-local, and cloud metadata (169.254.169.254) addresses. Enforce SSRF validation on every redirect hop (up to 3 hops).

3. Safely fetch HTML with response size limits (3 MB) and timeouts (8.0s), decompose non-content tags with BeautifulSoup, and extract clean paragraph text. If input is pasted text, validate length constraints (15 to 50,000 characters).

4. Extract 2 to 5 atomic, falsifiable claims using Google Gemini in JSON mode, validated into ExtractedClaim Pydantic models.

5. For each claim, execute parallel asynchronous queries against the Google Fact Check Tools API and the GNews API. Deduplicate retrieved evidence by canonical URL and normalized headline.

6. Pass claims and evidence to the verification engine; evaluate keyword overlap and publisher review ratings; classify each claim as Supported, Contradicted, Mixed, or Insufficient Evidence.

7. Calculate the deterministic credibility score:
   a. Compute normalized verdict balance as total directional impact divided by total claim weight.
   b. Calculate coverage ratio (evaluated claims / total claims) and evidence quality factor.
   c. Shift baseline (50.0) by verdict balance scaled by coverage, evidence quality, and publisher diversity bonuses.
   d. Clamp final score to [0, 100] and assign the corresponding rating label and confidence tier (High, Moderate, Low).

8. Invoke Gemini with constrained prompts to synthesize an objective editorial explanation grounded strictly in the retrieved evidence, retaining the backend-computed score.

9. Assemble the structured InvestigationReport JSON dossier and return HTTP 200.

10. Render the interactive report on the frontend, persist the case to the browser's localStorage ledger, and enable Markdown report export.

Note: If API keys are unconfigured and ENABLE_MOCK_FALLBACK=true, the pipeline generates deterministic mock data for offline testing.

End

BLOCK DIAGRAM

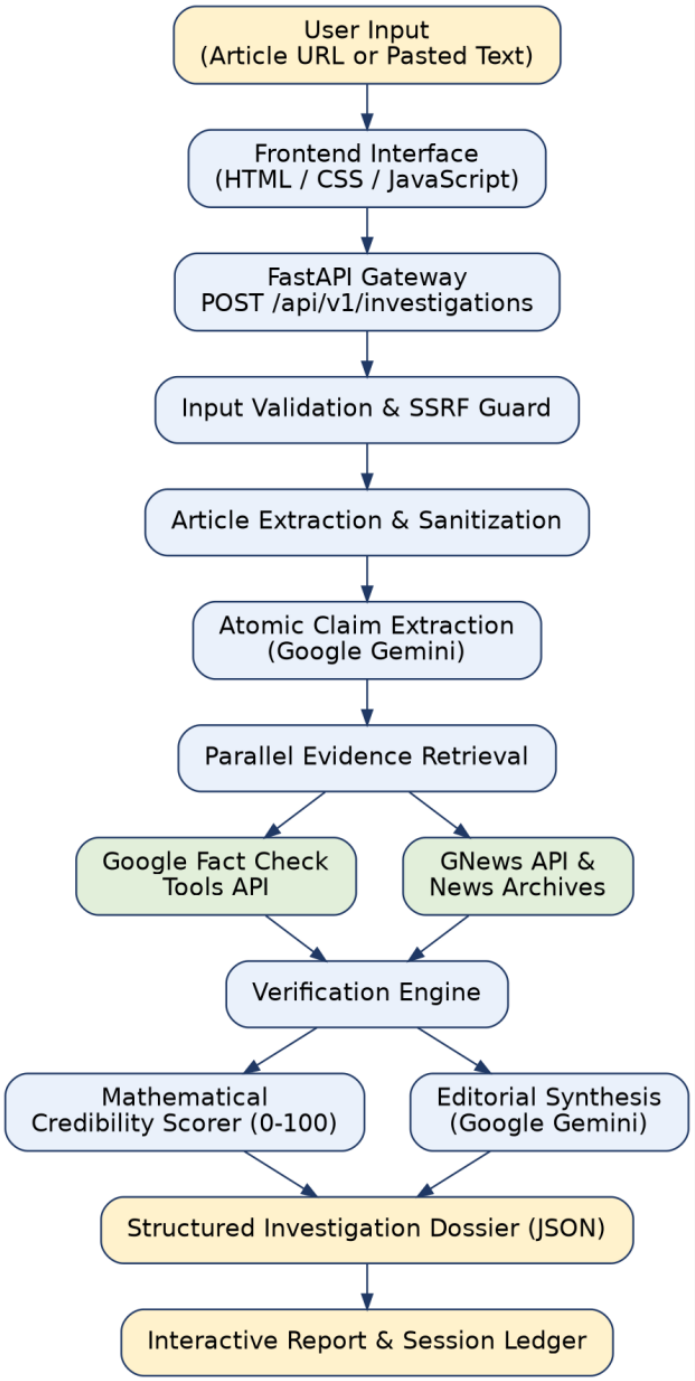


Fig. Block diagram of TruthLens

REQUIREMENT

Hardware Requirements

• Desktop or laptop computer with an active internet connection (or local offline evaluation)

• Minimum 4 GB RAM (8 GB recommended for concurrent browser development)

• Dual-core 2.0 GHz x86/64 processor or modern ARM64 processor

• Minimum 500 MB free disk storage for application and dependencies

Software Requirements

• Programming Language: Python 3.10 or higher (Python 3.13 tested and verified)

• Backend Web Framework: FastAPI 0.110+ with Uvicorn server

• Data Validation: Pydantic v2

• HTTP Client & Web Scraping: httpx 0.27+, BeautifulSoup4 4.12+

• Automated Testing: pytest 8.0+, pytest-asyncio 0.23+

• Frontend Technologies: Native HTML5, CSS3, JavaScript (Vanilla ES6 — zero npm dependencies)

• External APIs: Google Gemini API, GNews API, Google Fact Check Tools API (optional in mock mode)

• Supported Operating Systems: Microsoft Windows 10/11, macOS 12+, or Ubuntu/Debian Linux

• Supported Browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, or Apple Safari

ADVANTAGES

1. Evidence-Grounded Results: Conclusions are strictly anchored to verified fact-checks and news wires rather than speculative, hallucinated model outputs.

2. Deterministic & Explainable Scoring: The 0–100 credibility index is derived through an explicit mathematical balance formula rather than opaque, black-box classification.

3. Atomic Claim Decomposition: Complex news articles are decomposed into discrete factual assertions for isolated, granular cross-examination.

4. Asynchronous Concurrent Retrieval: Parallel querying across Google Fact Check Tools and GNews archives minimizes investigative latency.

5. Standardized 'Insufficient Evidence' Outcome: Prevents forced true/false speculation when authoritative documentation is missing in public registers.

6. Comprehensive SSRF Defenses: Enforces IP range validation, loopback blocking, cloud metadata protection, and redirect hop checks for safe article extraction.

7. Client-Side Auditable Ledger: Tracks all investigations locally in browser storage with instant search, filtering, and Markdown brief export.

8. Autonomous Mock Fallback Mode: Enables full end-to-end evaluation and offline demonstration without requiring paid external API credentials.

9. Verified Code Quality: Backed by 34 comprehensive asynchronous unit and integration tests achieving a 100% pass rate.

OUTPUT

For every investigation, the system returns a structured dossier which the frontend displays as an interactive report. The output contains:

• Overall Credibility Score: An index from 0 to 100 with its qualitative rating bracket (High Credibility, Mostly Credible, Mixed / Disputed Claims, Unverified / Insufficient Documentation, Low Credibility, or Fabricated) and formula confidence tier (High, Moderate, Low).

• Mathematical Formula Breakdown: Transparent disclosure of total claims, verdict counts, coverage ratio, verdict balance, and evidence quality factors.

• Atomic Claim Breakdown: List of atomic claims extracted from the submitted article or text with entity and category labels.

• Claim Verdicts & Rationales: Standardized verdict for each claim (Supported, Contradicted, Mixed, or Insufficient Evidence) with explicit evidentiary rationale and source citations.

• Catalog of Documented Evidence: Consulted source records categorized by type (Fact-Check or News Reporting) with publisher ratings and live URLs.

• Grounded Editorial Explanation: Structured synthesis containing an executive summary, key analytical findings, evidence synthesis, and documented investigative limitations.

• Session Ledger Record: Timestamped entry saved to browser localStorage, searchable and exportable as a Markdown verification brief (.md).

• Programmatic REST API: GET /api/v1/health (system and key status) and POST /api/v1/investigations (investigation execution).

LIMITATIONS

1. External API Dependencies & Rate Limits: Live verification relies on internet connectivity and the availability, quotas, and response times of the Gemini, GNews, and Google Fact Check APIs.

2. Bounded Evidentiary Coverage: Verification accuracy is limited by the public scope of indexed news archives and fact-checking registries; highly localized, niche, or newly emergent events may return Insufficient Evidence.

3. Language Model Sensitivity: Claim extraction and editorial summaries depend on the generative quality, prompt adherence, and neutrality of the underlying language model.

4. Rule-Based Mathematical Scoring: The credibility score is calculated by a deterministic, rule-based formula designed to measure external documentation; it measures evidentiary support rather than absolute philosophical truth.

5. Browser-Local Storage Scope: The investigation ledger is persisted strictly within the client's browser localStorage, meaning case histories do not automatically synchronize across different devices or browsers without manual Markdown export.

APPLICATIONS

1. Newsrooms and Journalism: Provides rapid, evidence-grounded cross-referencing for journalists vetting breaking stories, wire reports, and press releases.

2. Independent Fact-Checking Desks: Serves as an automated triage tool to identify atomic claims within lengthy articles and match them against existing registries.

3. Media Literacy Education: Practical educational instrument in universities and schools to teach students how claims are decomposed, cross-referenced, and audited.

4. General User Claim Auditing: Allows everyday news readers to verify suspicious social media posts, forwards, and sensational articles before sharing.

5. Misinformation Research: Provides researchers with an open, reproducible framework for studying claim-evidence relationships and verification pipelines.

6. Compliance and Editorial Archives: Delivers auditable verification briefs (.md) for organizations requiring evidentiary paper trails for published content.

CONCLUSION

TruthLens provides a transparent, explainable, and practical system for automated news and claim verification. By combining LLM-based claim extraction with concurrent retrieval from registered fact-checkers and news archives, and by using a deterministic scoring formula, it provides users with an auditable credibility evaluation grounded in verifiable records.

The system's strict architectural safeguards—including comprehensive anti-SSRF protections, honest reporting of Insufficient Evidence, client-side session ledger persistence, and 34 automated unit and integration tests—make it a robust, dependable engineering solution to combat digital misinformation.

By replacing speculative AI opinions with structured, source-attributed verification dossiers, TruthLens demonstrates how artificial intelligence and deterministic scoring can be paired effectively to foster media integrity and informational transparency.

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